

Databricks in Azure: from sandbox to security review

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Intro

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03 When production says ‘hello’ (security has entered the chat)

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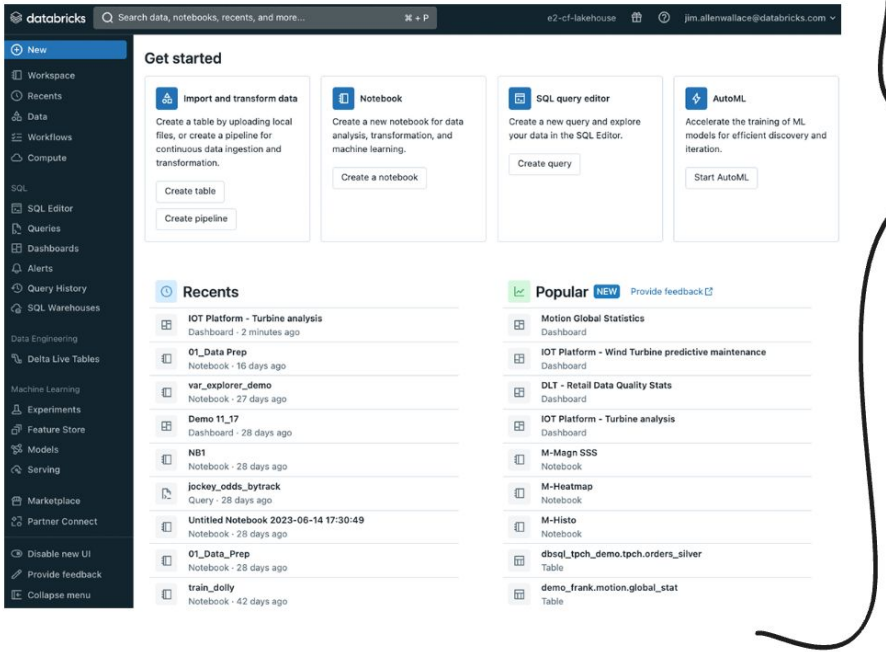
LinkedIn



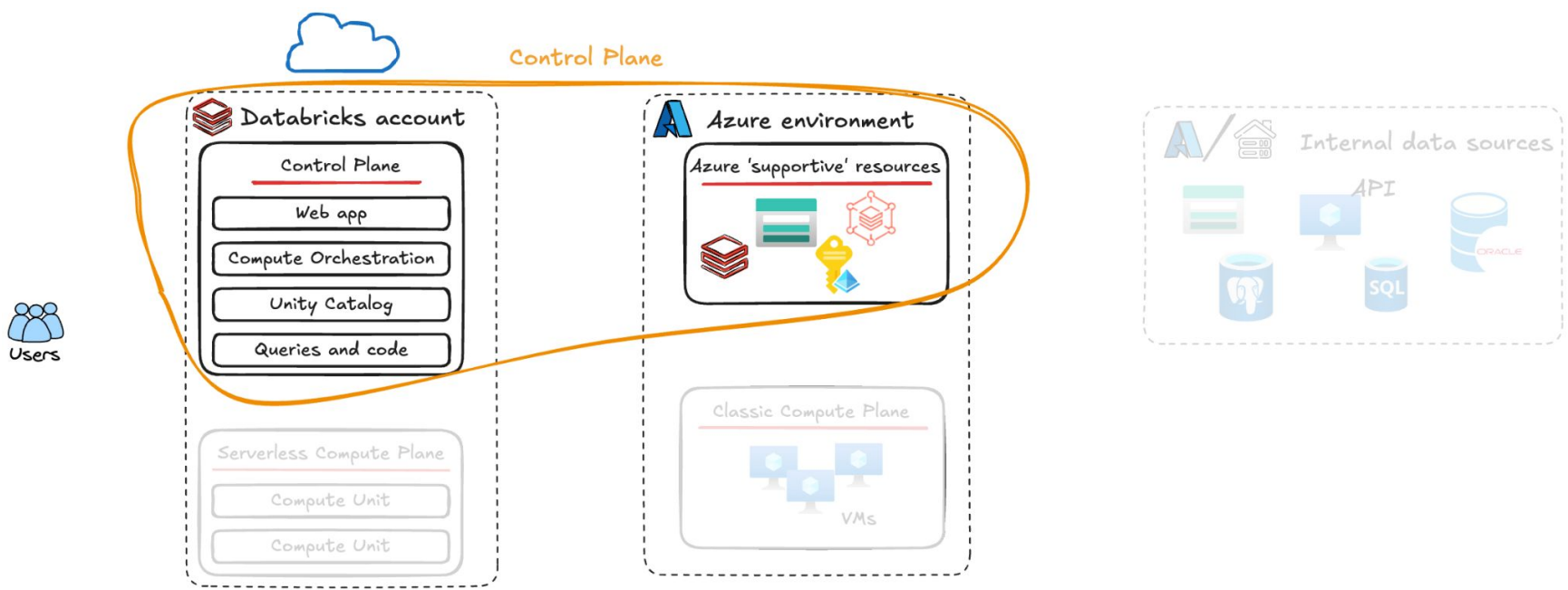
Husqvarna

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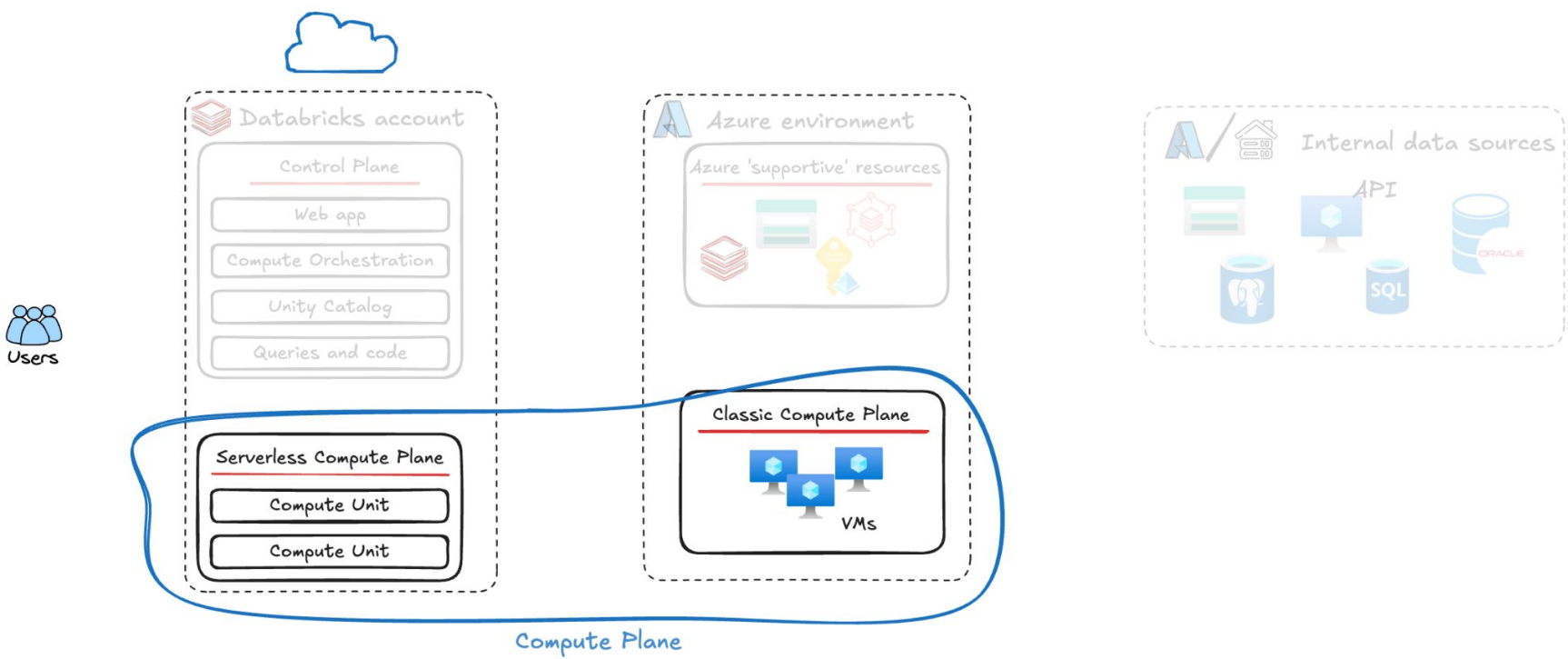
Databricks in Azure? Give me 15 minutes



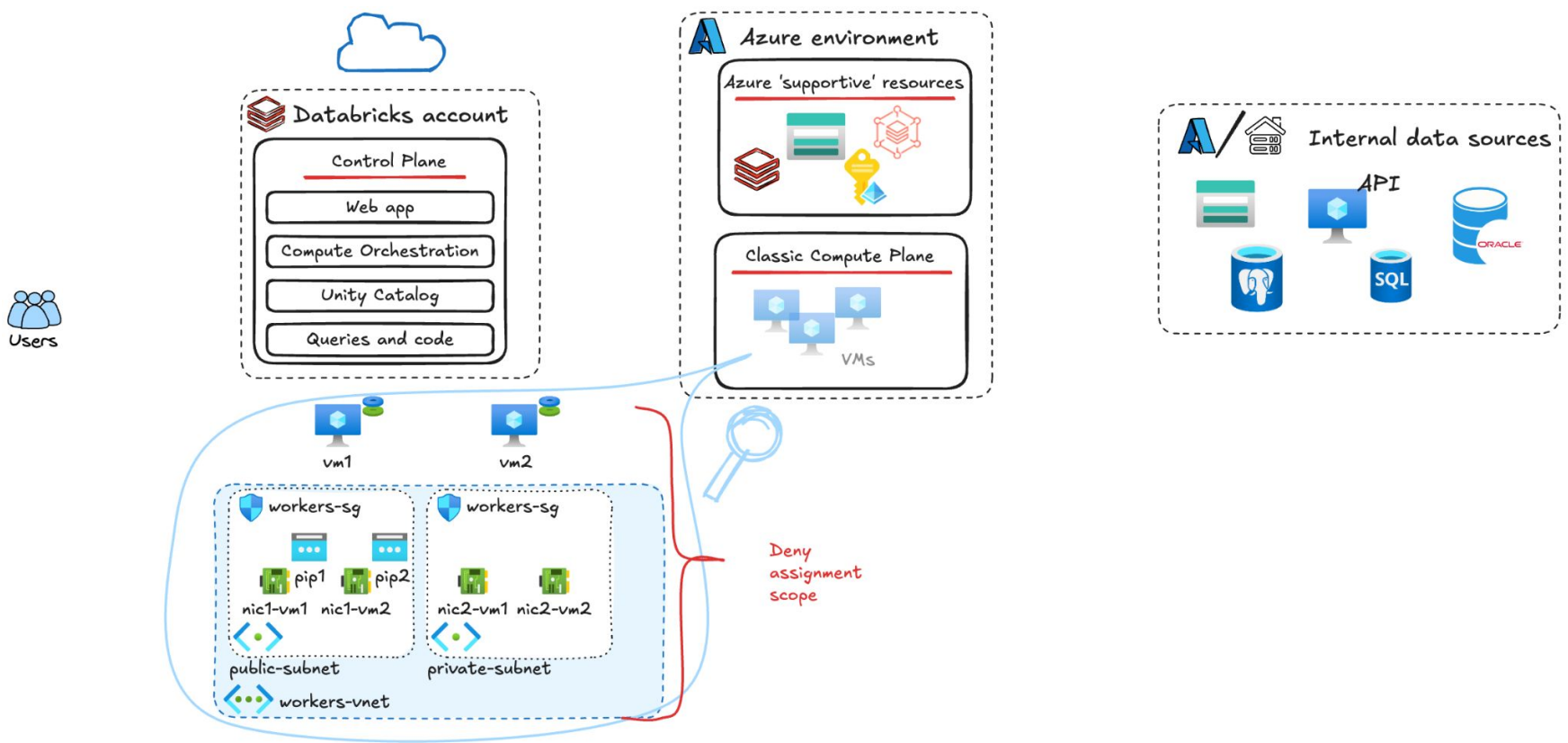
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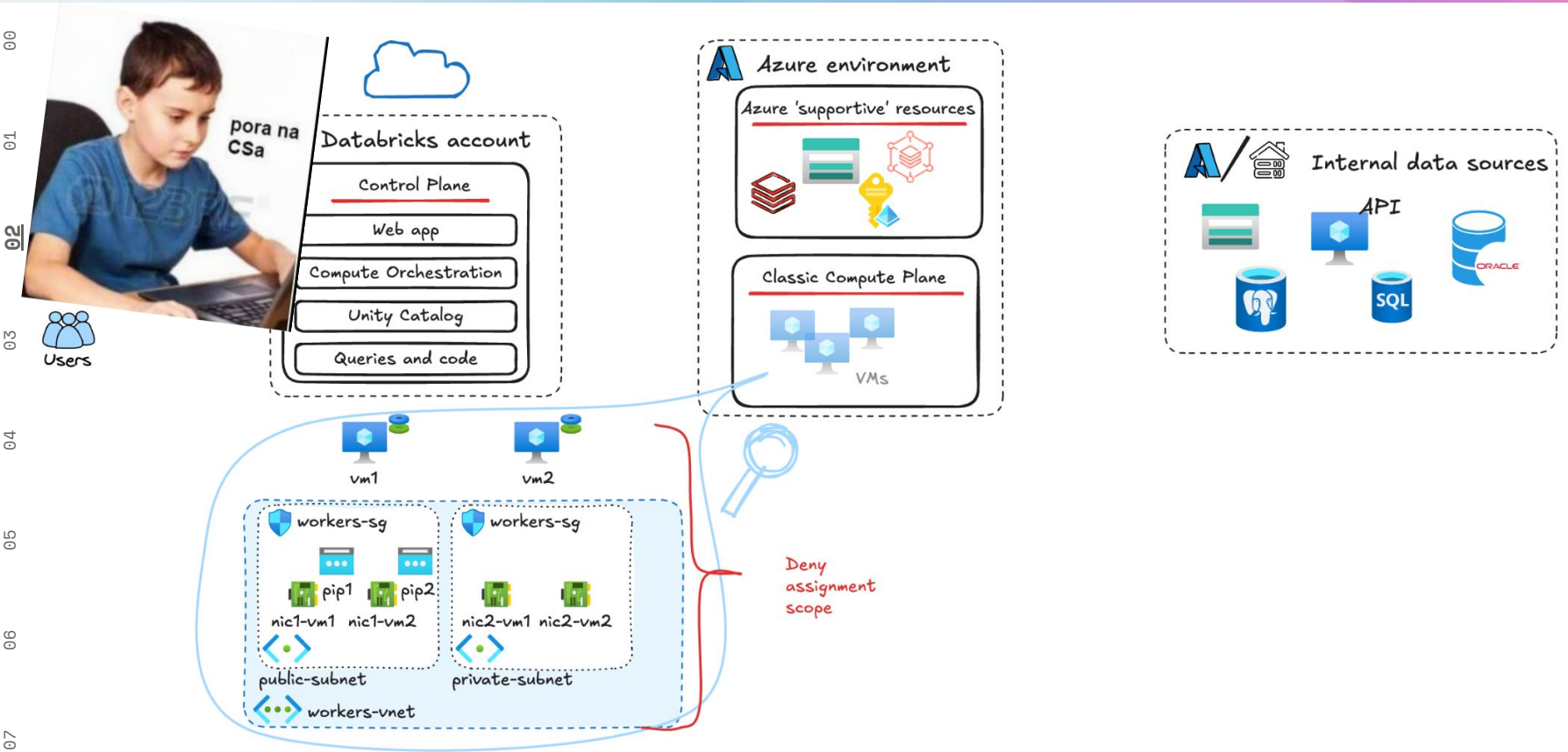
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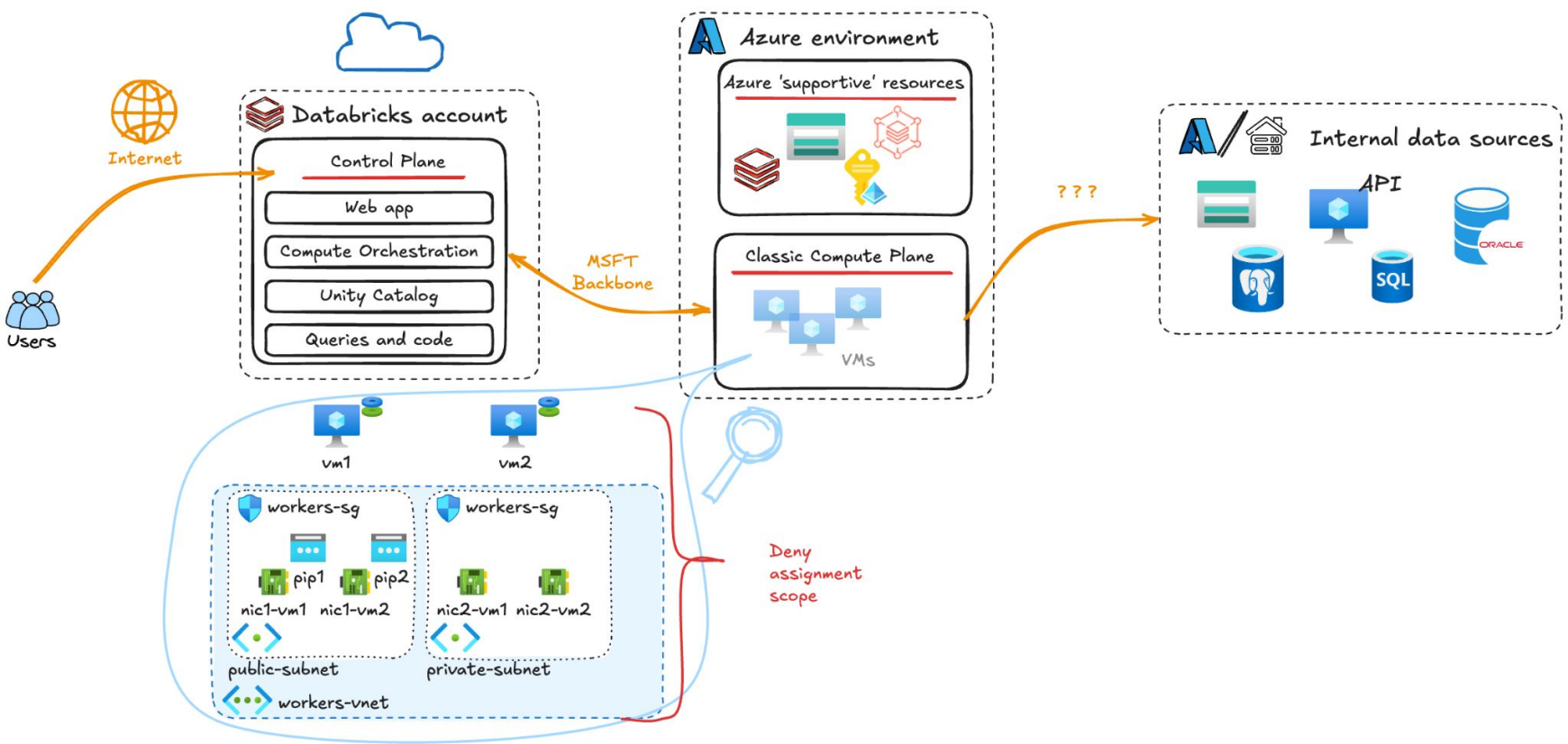
My super cool sandbox environment



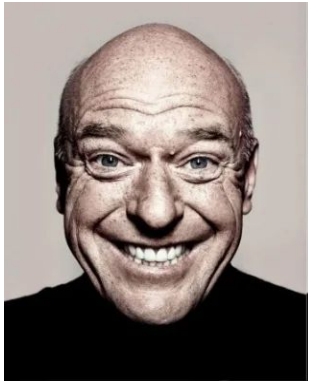
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My super cool sandbox environment

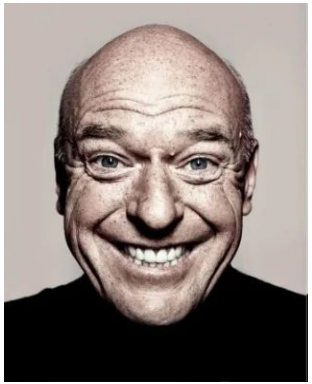


When production says ‘hello’ (security has entered the chat)

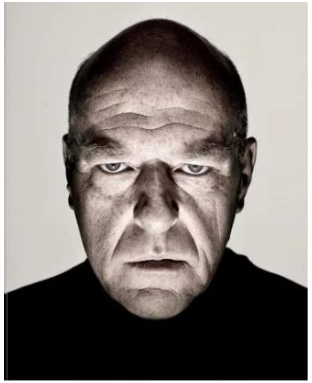


Me with my deployment

When production says ‘hello’ (security has entered the chat)



Me with my deployment



Me after security review

Best practices for security, compliance, and privacy

Summarize this article for me

The Databricks Security Best Practices guide, including a downloadable PDF, is available on the Databricks Security & Trust Center. The sections in this article list the best practices from this guide along the principles of this pillar.

1. Manage identity and access using least privilege

Account setup and identity configuration

During deployment, configure Databricks account administration, SSO, and user provisioning to establish a secure foundation:

- Assign account admin roles to 2-3 trusted individuals only
- Configure SSO with Azure Active Directory (Azure AD) using OIDC or SAML

The “It’s just a few checkboxes” phase

Network deployment considerations for Azure

Deploy secure network infrastructure for Databricks workspaces on Azure. The following steps establish secure connectivity:

- Create a VNet with minimum /18 address space for workspace deployments
- Provision public and private subnets delegated to `Microsoft.Databricks/workspaces`
- Configure Network Security Groups (NSGs) to control traffic (Databricks manages default NSG rules)
- Deploy Azure Private Link for private connectivity to Databricks control plane and web application
- Enable Secure Cluster Connectivity (No Public IP) to eliminate public IP addresses on cluster nodes
- Configure VPN Gateway or ExpressRoute for on-premises connectivity (if required) -> platform
- Implement network segmentation to isolate production and non-production environments

VNet Injection

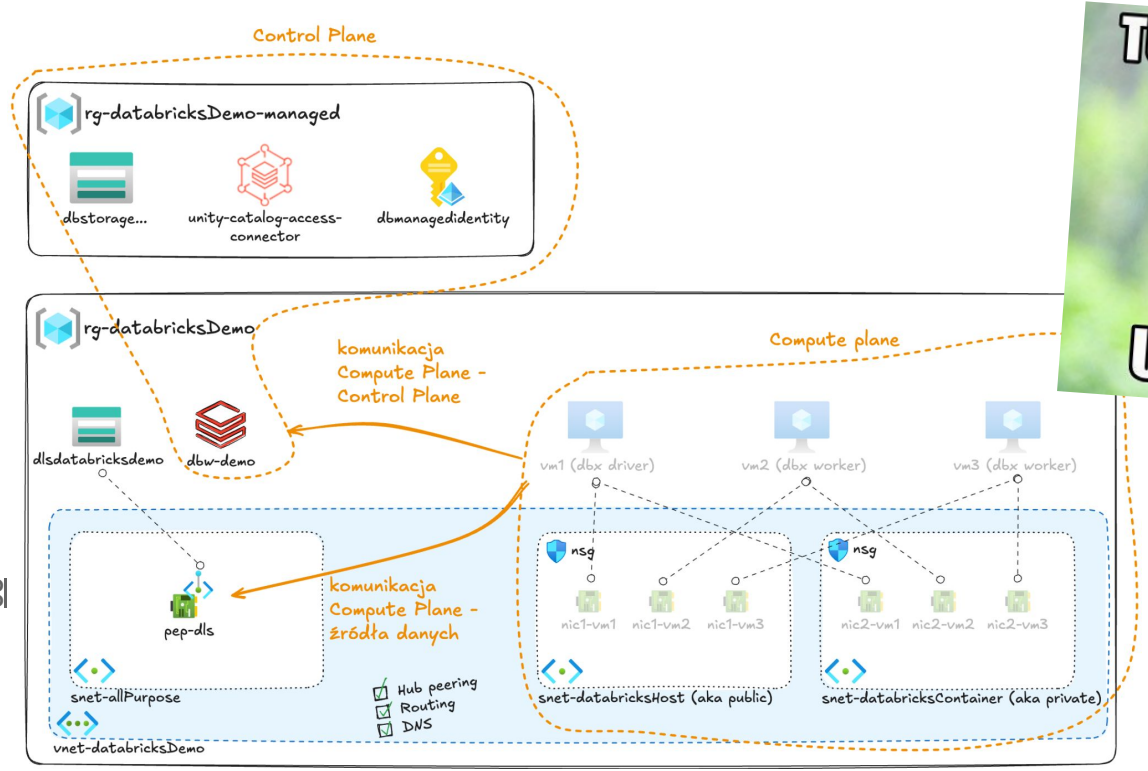
SCC/ NPIP

For step-by-step Azure network configuration, see Azure network architecture.

Network security best practices

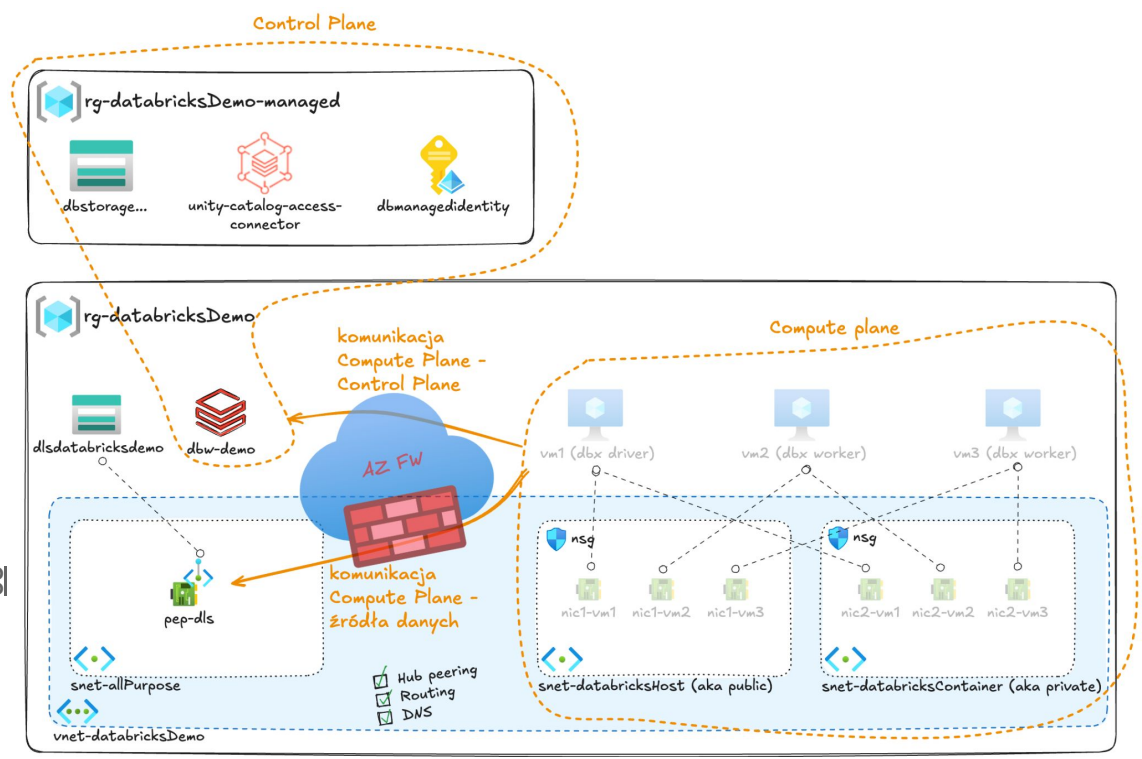
- Use Secure Cluster Connectivity (No Public IP)
- Deploy Azure Databricks into your own Azure virtual network -> VNet Injection
- Configure IP access lists
- Use Azure PrivateLink
- Implement network exfiltration protections
- Isolate Azure Databricks workspaces into different networks
- Configure a firewall for serverless compute access
- Restrict access to valuable codebases to only trusted networks
- Use Virtual network encryption

Understanding the Databricks beast

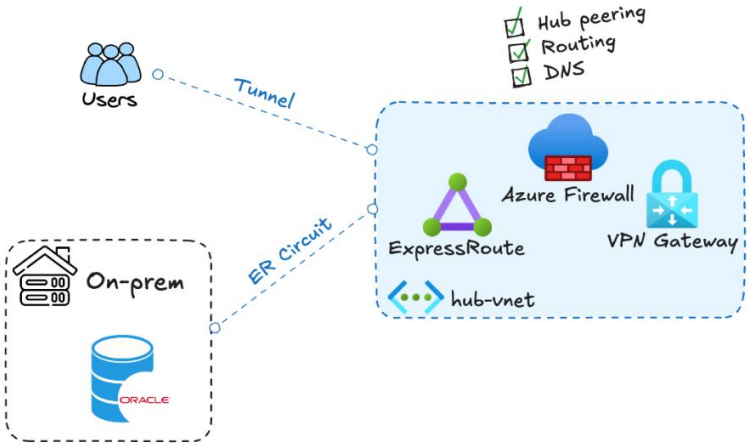




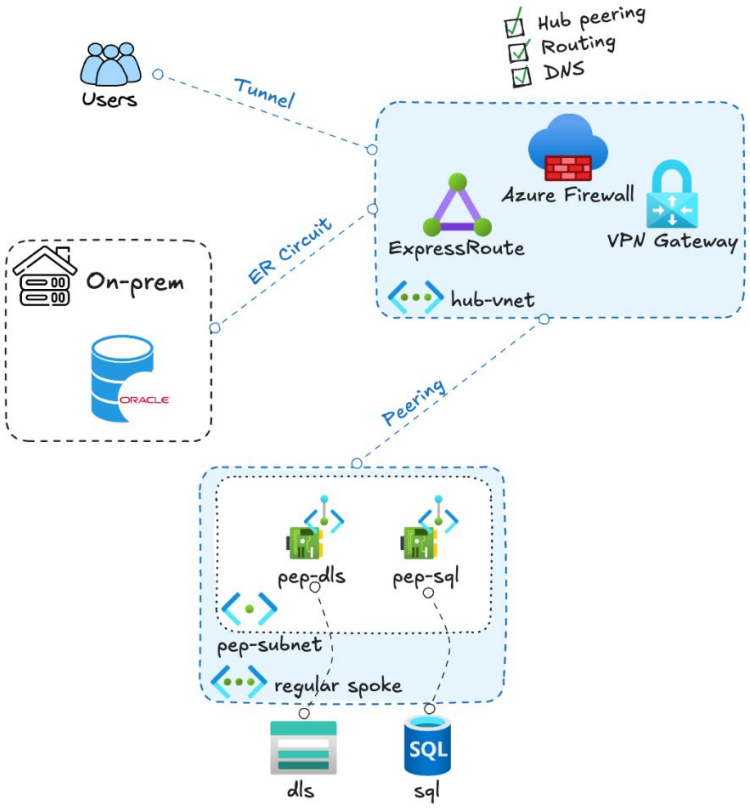
Understanding the Databricks beast



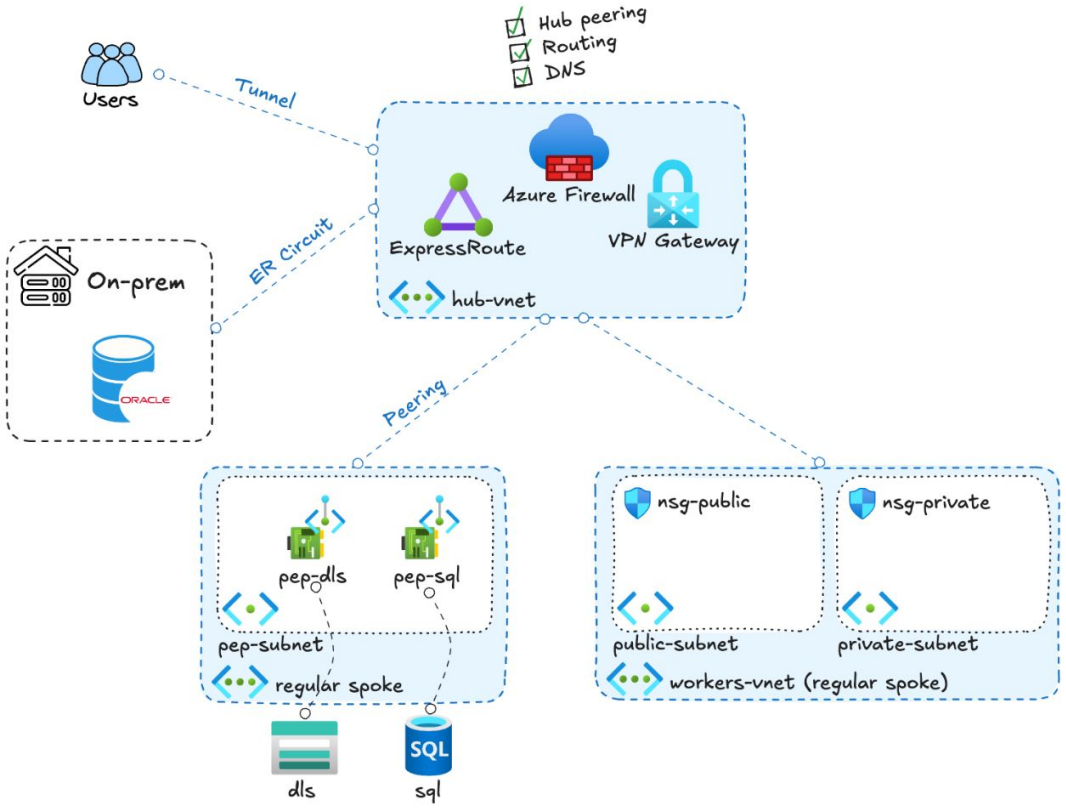
Let’s build it better this time



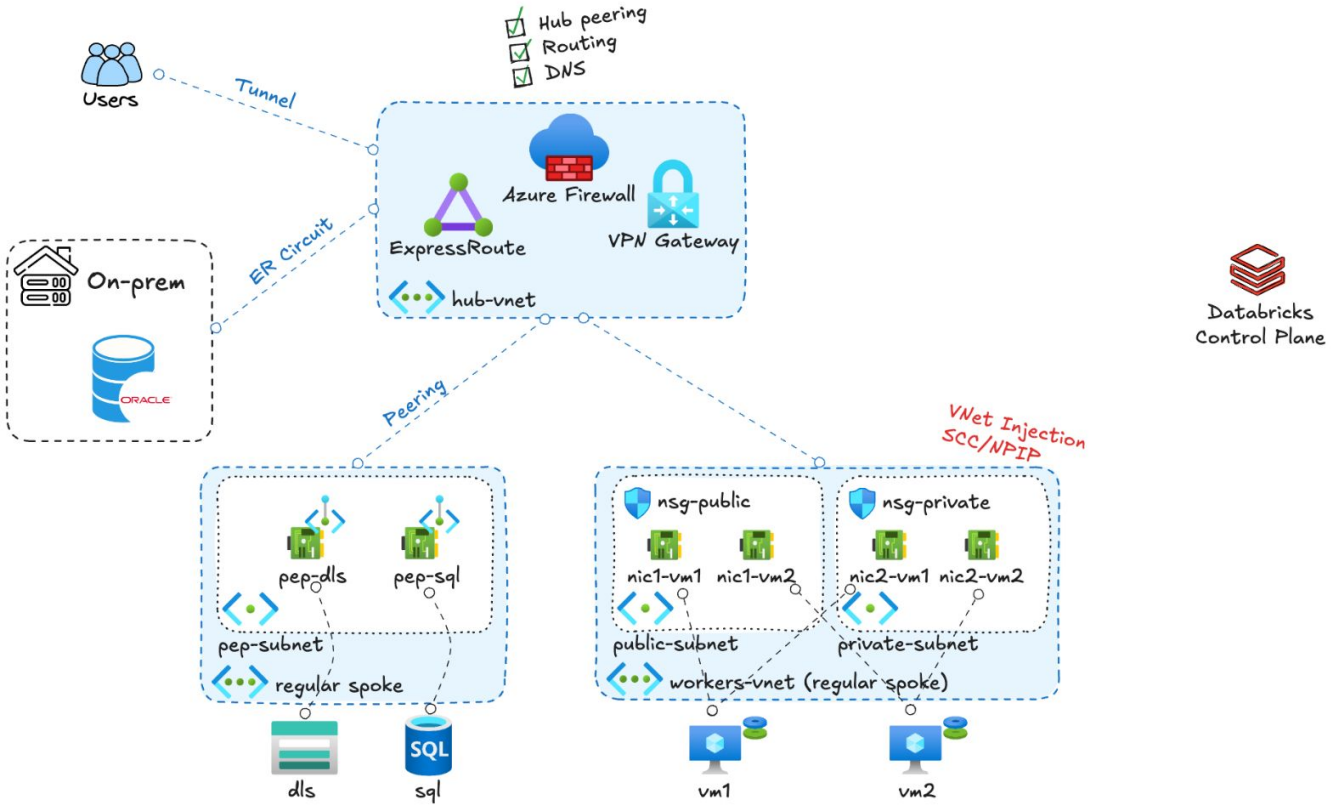
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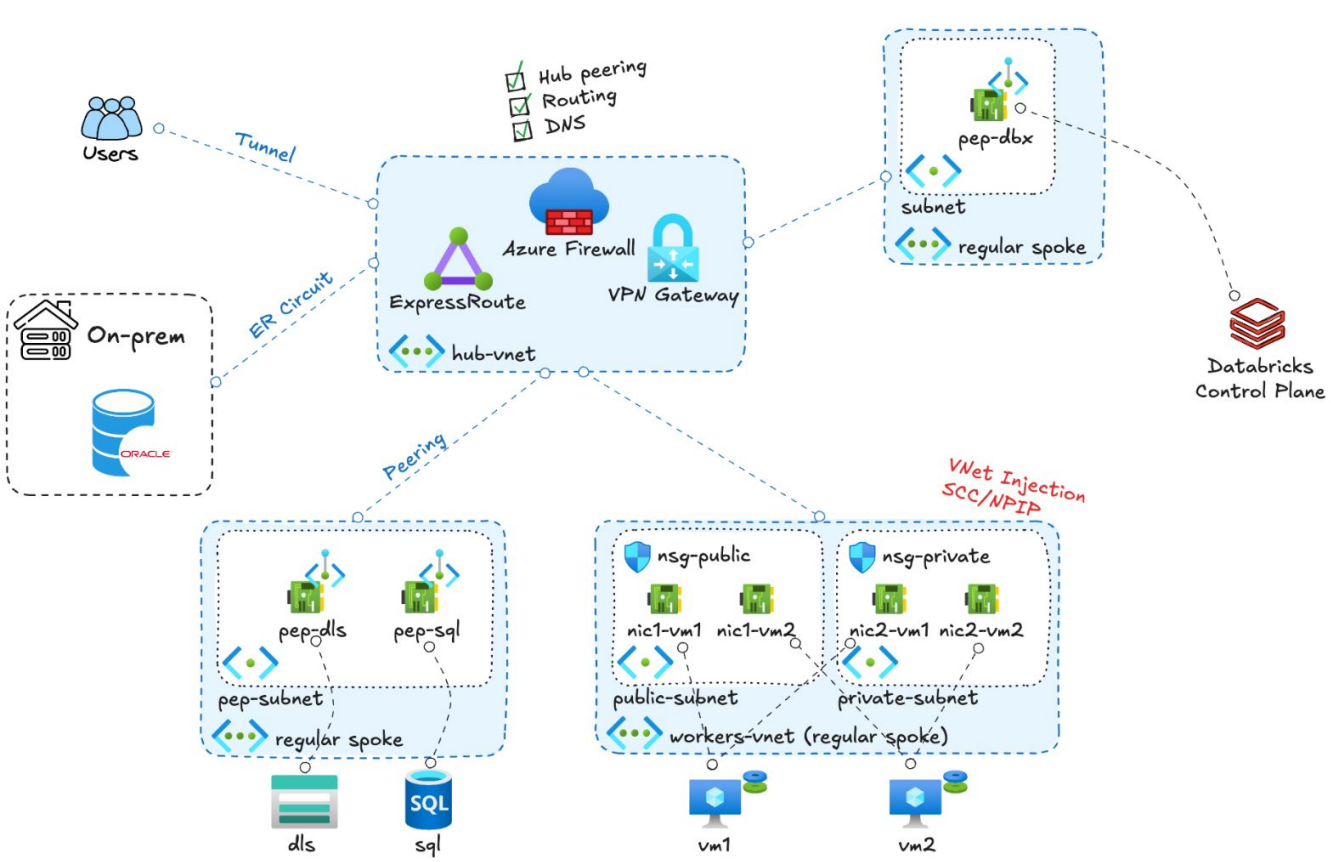
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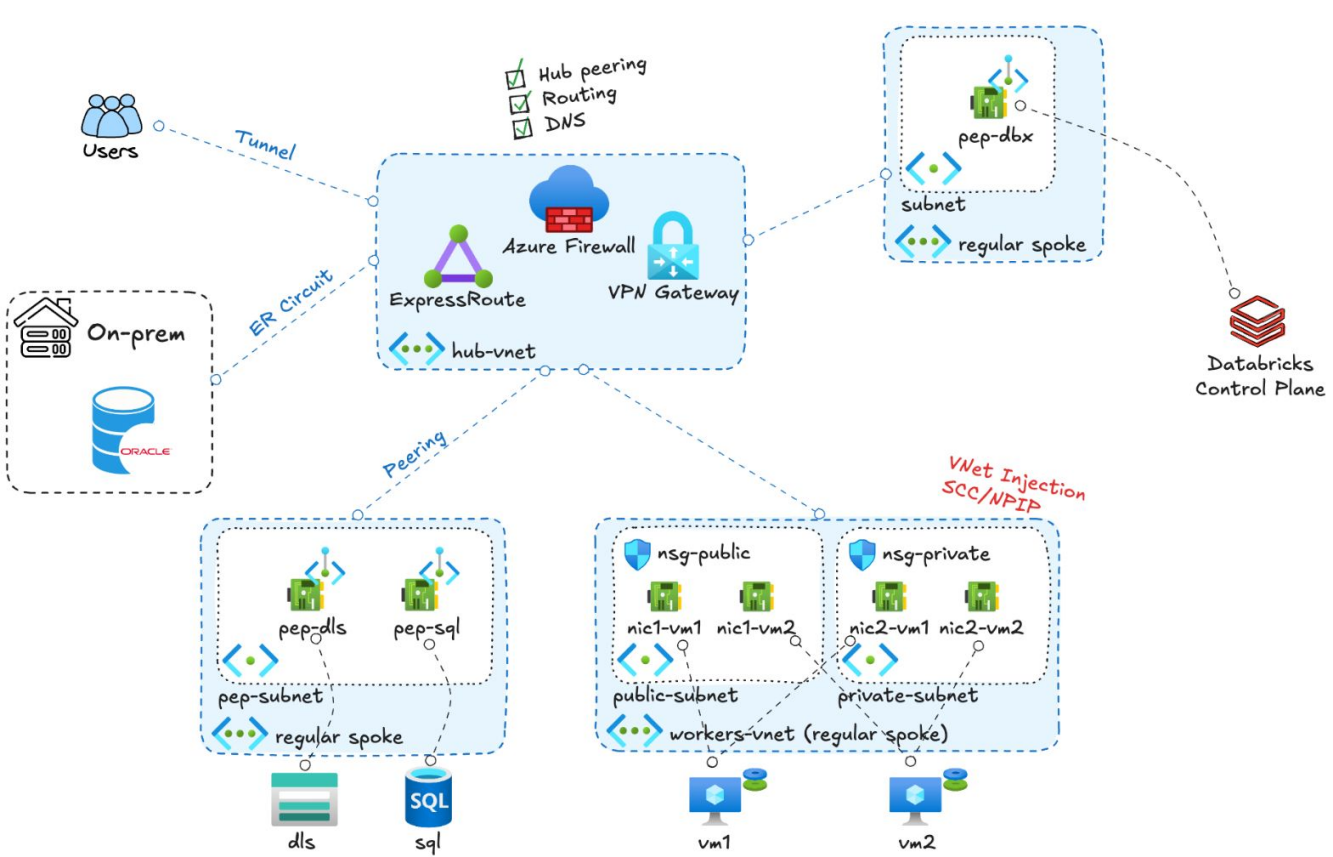
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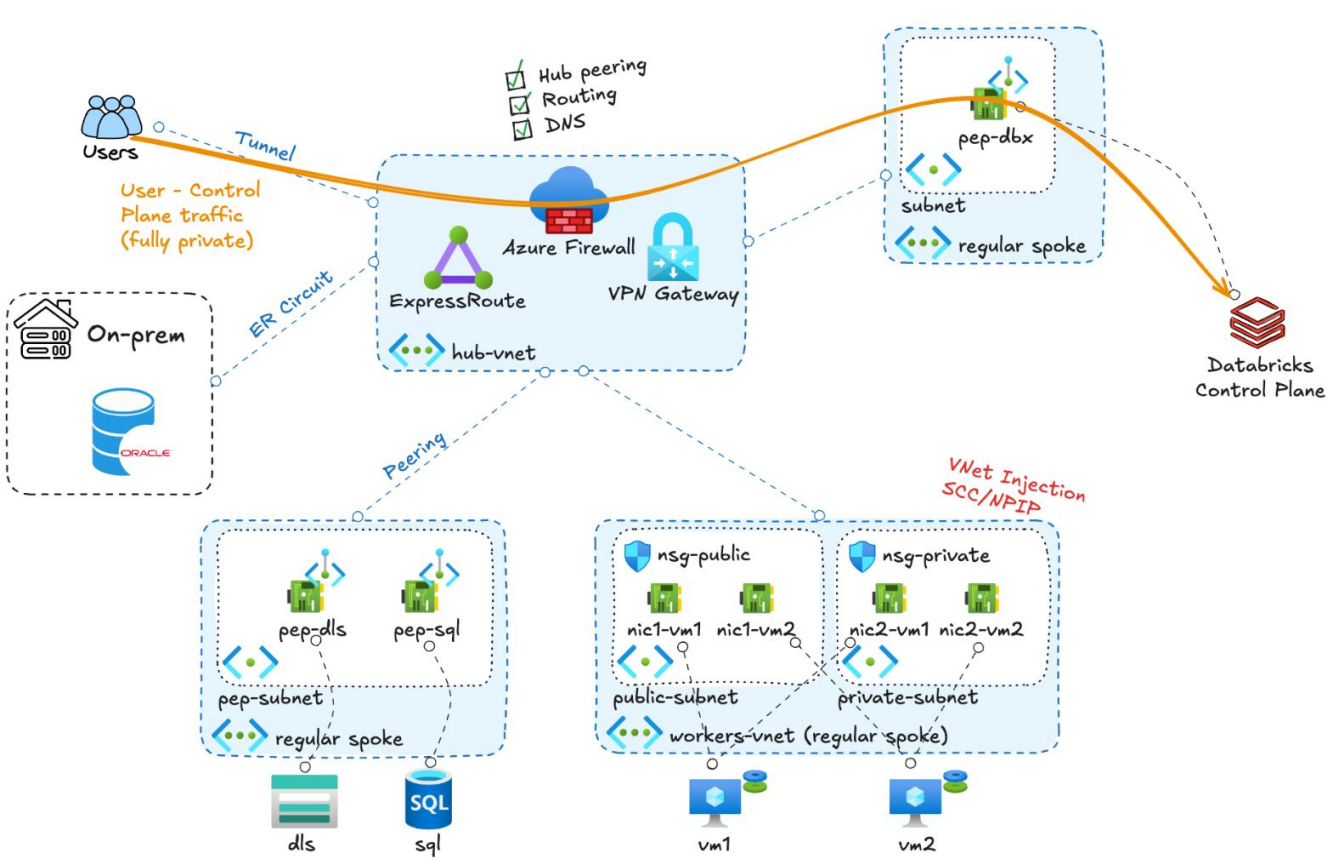


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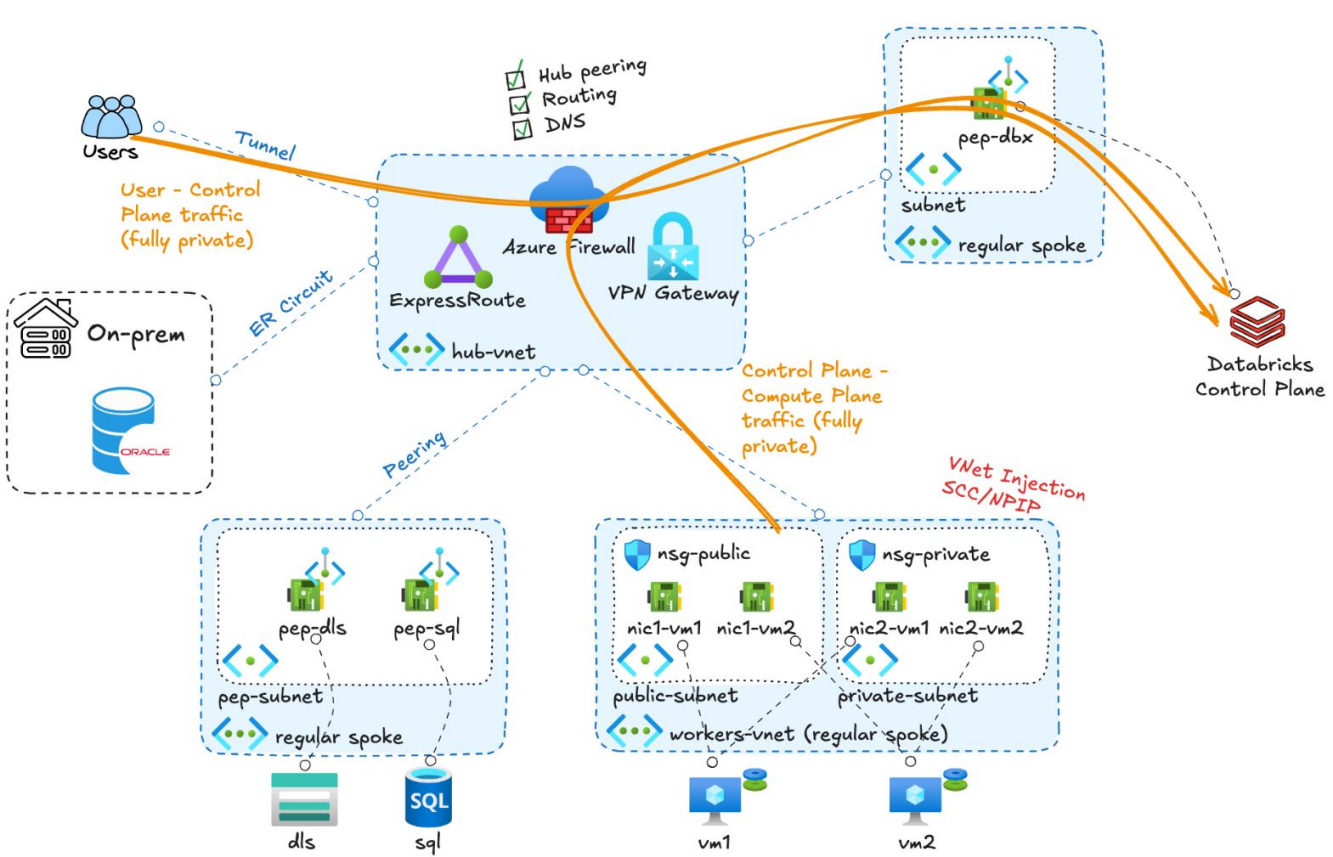


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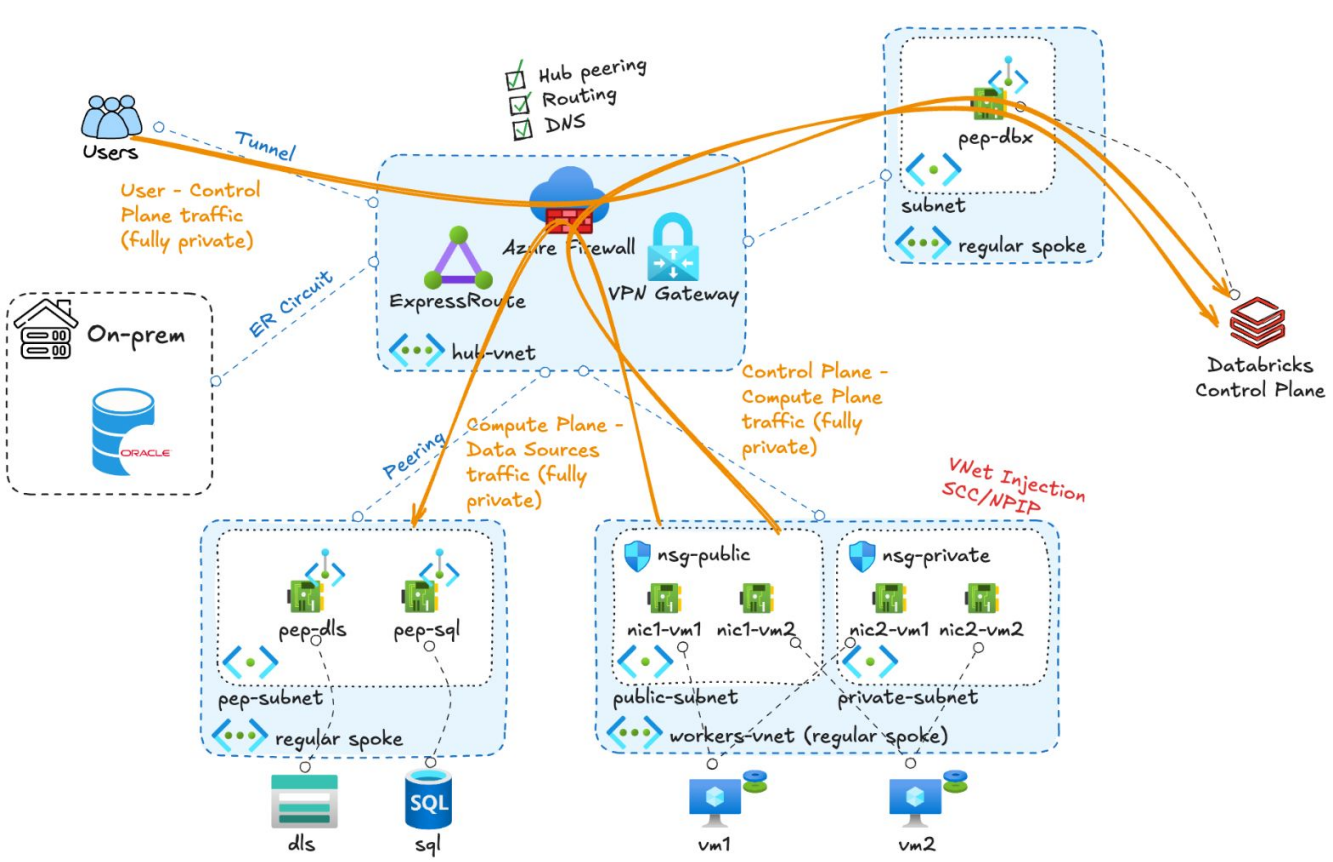


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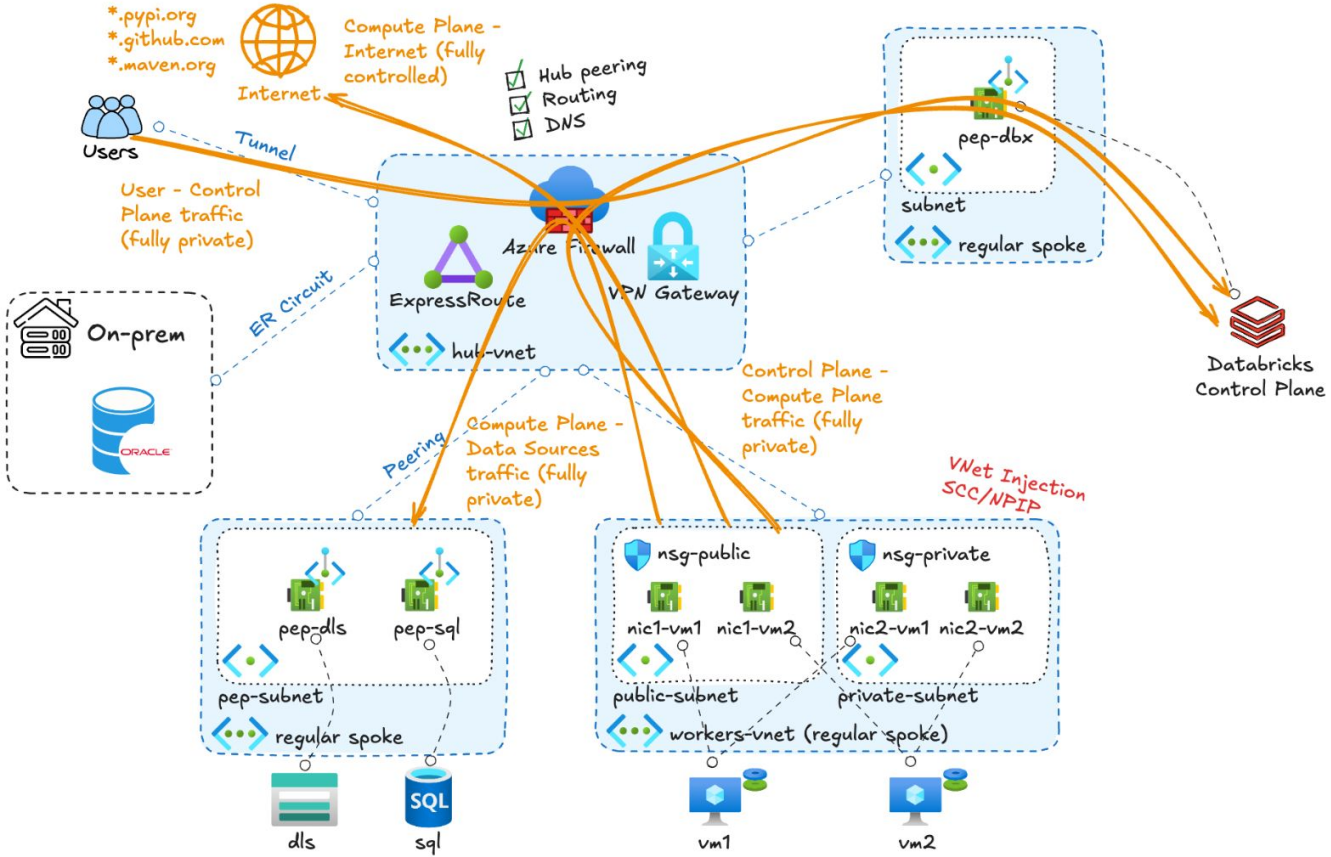


Let’s build it better this time





Let’s build it better this time





Let's build it better this time



Firewall Closer look:

dbx public subnet

dbx pep

dls pep

different ports

	SourceIP	DestIP	DestPort	Protocol	Action	Rule
:6 13:12:28	10.65.182.5	10.65.105.140	3306	TCP	Allow	allow-databricks0001-to-DatabricksPEP
:6 13:12:28	10.65.182.5	10.65.105.140	3306	TCP	Allow	allow-databricks0001-to-DatabricksPEP
:6 13:10:13	10.65.182.5	10.65.91.10	443	TCP	Allow	allow-databricks0001-to-Datasource
:6 13:10:12	10.65.182.5	10.65.105.140	8443	TCP	Allow	allow-databricks0001-to-DatabricksPEP
:6 13:09:26	10.65.182.4	10.65.105.140	8444	TCP	Allow	allow-databricks0001-to-DatabricksPEP
:6 13:08:45	10.65.182.5	10.65.105.140	8444	TCP	Allow	allow-databricks0001-to-DatabricksPEP
:6 13:08:40	10.65.182.4	10.65.105.140	8445	TCP	Allow	allow-databricks0001-to-DatabricksPEP
:6 13:08:40	10.65.182.4	10.65.105.140	443	TCP	Allow	allow-databricks0001-to-DatabricksPEP
:6 13:08:40	10.65.182.4	10.65.105.140	3306	TCP	Allow	allow-databricks0001-to-DatabricksPEP
:6 13:08:38	10.65.182.4	10.65.105.140	8443	TCP	Allow	allow-databricks0001-to-DatabricksPEP
:6 13:08:38	10.65.182.4	10.65.105.140	8443	TCP	Allow	allow-databricks0001-to-DatabricksPEP
:6 13:08:37	10.65.182.4	10.65.105.140	8444	TCP	Allow	allow-databricks0001-to-DatabricksPEP
:6 13:08:37	10.65.182.4	10.65.105.140	8443	TCP	Allow	allow-databricks0001-to-DatabricksPEP

Conclusions: what we wish we knew from day one

00

01 "Secure by default" ≠ secure enough

02 Be as secure as you need to be

03 Databricks lives between data and infra

04 Design for security from day one

05 VNet-injected Databricks is just another spoke

06 Reduce the problem to VM → endpoint connectivity

07

Resources

Classic Compute Plane connectivity:
https://learn.microsoft.com/en-us/azure/databricks/security/network/classic/private-link-standard

VNet injection:
https://docs.azure.cn/en-us/databricks/security/network/classic/vnet-inject

Azure Databricks - best practices for security, compliance and privacy:
https://learn.microsoft.com/en-us/azure/databricks/lakehouse-architecture/security-compliance-and-privacy/best-practices

Questions time

